

Partial Differential Equations PhD Exam, May 2005

1. Let Ω be a bounded smooth set in $\mathbb{R}^n$ ($n \geq 2$) and let $L : \mathbb{R}^n \times \mathbb{R} \times \overline{\Omega} \rightarrow \mathbb{R}$ be a smooth mapping. Assume in addition that L is bounded below and the mapping $p \mapsto L(p, z, x)$ is convex for each $z \in \mathbb{R}$, $x \in \overline{\Omega} \subseteq \mathbb{R}^n$. Prove that $I(w) := \int_{\Omega} L(Dw, w, x) dx$ is weakly lower semicontinuous on $W^{1,q}(\Omega)$ ($1 < q < \infty$).
2. Let $\{a_{ij}\}_{n \times n}$ ($n \geq 2$) be a positive definite constant matrix. Let B_1 be the unit ball in $\mathbb{R}^n$. Find the Euler–Lagrange equation for

$$I(w) = \int_{B_1} \sum_{i,j=1}^n (a_{ij} \partial_i w \partial_j w) dx, \quad w \in H_0^1(B_1),$$

under the constraint $\int_{B_1} |w|^2 = 1$.

3. Prove that if $\lambda > 0$ is large enough, there exists a function $u \in H^2(B_1) \cap H_0^1(B_1)$ solving

$$\begin{cases} -\Delta u + \lambda u = \sqrt{1 + |Du|^2}, & \text{in } B_1, \\ u = 0, & \text{on } \partial B_1. \end{cases}$$

4. Let Ω be a bounded smooth subset of $\mathbb{R}^n$ ($n \geq 2$) and let u be a classical solution of

$$\begin{cases} \Delta u(x) + a(x)u(x) = 0, & \Omega, \\ u \geq 0, & \partial\Omega, \end{cases}$$

where $a(x)$ is a smooth function. Suppose there exists $\phi(x) > 0$ and smooth over $\overline{\Omega}$ such that

$$\Delta \phi + a(x)\phi \leq 0, \quad \Omega.$$

Show that either $u > 0$ or $u = 0$ in Ω .

5. Let $u \in C^2(B_1)$ be a solution of

$$\begin{cases} -\Delta u = e^u, & B_1 \subset \mathbb{R}^2, \\ u = 1, & \text{on } \partial B_1. \end{cases}$$

Prove that:

(1): $u > 1$ in B_1 .

(2): u is radially symmetric and $\frac{du}{dr} < 0$ for all $r \in (0, 1)$.

6. This problem has two parts.

(a) Show there exists a unique minimizer $u \in A$ of

$$I[w] = \int_{\Omega} \left(\frac{1}{2} |Dw|^2 - fw \right) dx,$$

where Ω is a bounded smooth subset of $\mathbb{R}^n$ ($n \geq 2$), $f \in L^2(\Omega)$,

$$A = \{w \in H_0^1(\Omega); |Dw| \leq 1 \text{ a.e.}\}.$$

(b) Prove

$$\int_{\Omega} Du \cdot D(w - u) dx \geq \int_{\Omega} f(w - u) dx$$

for all $w \in A$.

7. Let $u \in W^{1,p}(B_1)$. Prove that $u^+, u^-, |u| \in W^{1,p}(B_1)$ and for almost all points over each region ($u > 0, u < 0, u = 0$) (Recall $u = u^+ - u^-$),

$$Du^+ = \begin{cases} Du, & u > 0, \\ 0, & u \leq 0, \end{cases} \quad Du^- = \begin{cases} -Du, & u < 0, \\ 0, & u \geq 0, \end{cases} \quad D|u| = \begin{cases} Du, & u > 0, \\ -Du, & u < 0, \\ 0, & u = 0. \end{cases}$$

Hint: Let

$$f_{\epsilon}(u) = \begin{cases} (u^2 + \epsilon^2)^{1/2} - \epsilon, & u > 0, \\ 0, & u \leq 0. \end{cases}$$

For all $\phi \in C_0^1(B_1)$,

$$\int_{B_1} f_{\epsilon}(u) D\phi dx = - \int_{\{u>0\}} \frac{u Du}{\sqrt{u^2 + \epsilon^2}} \phi dx.$$

8. Let $a_{ij}(x)$, $b_i(x)$ ($i, j = 1, \dots, n$), $c(x)$ be C^∞ functions over B_1 ($n \geq 2$). Let λ_0, λ_1 be two positive constants satisfying

$$0 < \lambda_0 |\xi|^2 \leq a_{ij}(x) \xi_i \xi_j \leq \lambda_1 |\xi|^2 \quad \forall x \in B_1, \quad \forall \xi \in \mathbb{R}^n.$$

Suppose $u \in W^{1,q}$ is a weak solution of

$$\sum_{i,j} \partial_i (a_{ij}(x) \partial_j u) + \sum_i b_i(x) \partial_i u(x) + c(x) u = u^2, \quad B_1.$$

Look for the smallest $q_0 > 1$ such that for $q > q_0$, $u \in C^\infty(B_1)$. Prove your statement.

Hint: The following estimates should be used: Suppose $u \in W^{1,q}(B_1)$ verifies

$$\sum_{i,j} \partial_i (a_{ij}(x) \partial_j u) + \sum_i b_i(x) \partial_i u(x) + c(x) u = f(x), \quad B_1,$$

in the sense of distribution, $f \in L^q$, $q \in (1, \infty)$. Then

$$\|u\|_{W^{2,q}(B_{1/2})} \leq C (\|u\|_{L^q(B_1)} + \|f\|_{L^q(B_1)}).$$

If we further know $u \in C^\alpha(\overline{B_1})$ and $f \in C^\alpha(\overline{B_1})$ for some $\alpha \in (0, 1)$, then the following Schauder estimate holds:

$$\|u\|_{C^{2,\alpha}(B_{1/2})} \leq C (\|u\|_{C^\alpha(B_1)} + \|f\|_{C^\alpha(B_1)}).$$